

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - Traversal Challenge

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.		
Student Name:	K.KAVIYA	Reg. No.:	192521262
Section / Batch:	1	Date:	31/07/2026

Code of Conduct

I K.KAVIYA(192521262) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.KAVIYA

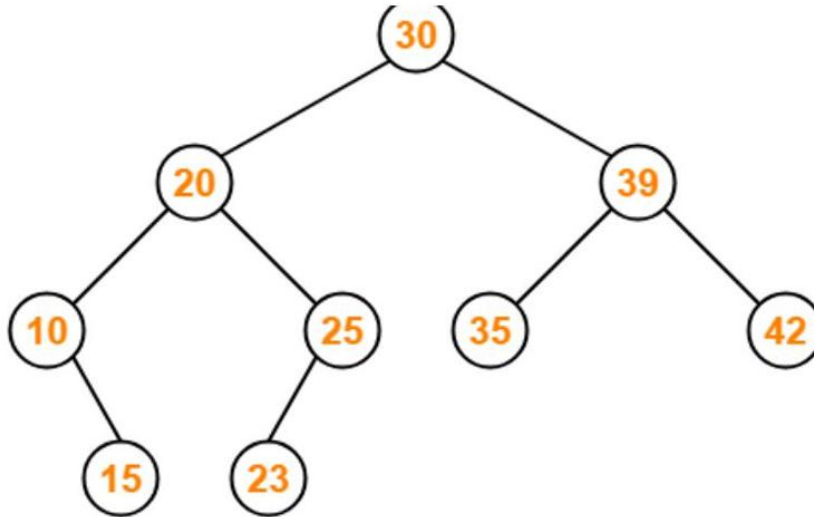
Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:			100 Marks

Question No:1

Write the traversal sequences for following trees



Preorder

Root → Left → Right

30, 20, 10, 15, 25, 23, 39, 35, 42

Inorder

Left → Root → Right

10, 15, 20, 23, 25, 30, 35, 39, 42

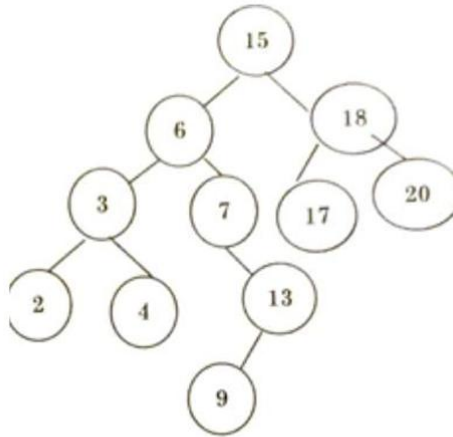
Postorder

Left → Right → Root

15, 10, 23, 25, 20, 35, 42, 39, 30

Question No: 2

Write the Inorder, Preorder & Postorder traversal sequences for following trees

**Preorder**

Root → Left → Right

15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

Inorder

Left → Root → Right

2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20

Postorder

Left → Right → Root

2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

The Q2 diagram is on page 1 of your PDF.

Question No: 3

Consider the following Inorder and Preorder traversal of a tree

Inorder: D B E F A G H C

Preorder: A B D E F C G H

What is the Postorder traversal of the same?

1. D F E B H C G A

2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

Inorder:

D B E F A G H C

Preorder:

A B D E F C G H

The question asks for Postorder.

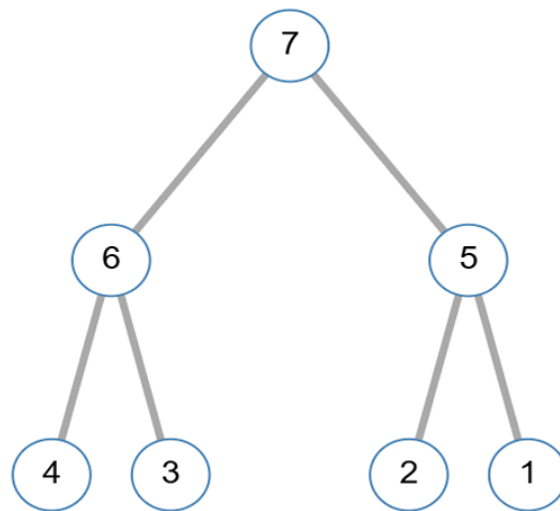
Answer:

D F E B H G C A

So the correct option is **Option 3**.

Question No: 4

Find Inorder, Preorder and Postorder



Preorder

Root → Left → Right

7, 6, 4, 3, 5, 2, 1

Inorder

Left → Root → Right

4, 6, 3, 7, 2, 5, 1

Postorder

Left → Right → Root

4, 3, 6, 2, 1, 5, 7

Q5. Construct the BST

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

Numbers are inserted in this order:

7, 5, 1, 8, 3, 6, 0, 9, 4, 2

The question specifies a standard BST ordering.

Step 1: Construct the tree

```
      7
     /\
    5  8
   /\  \
  1 6  9
 /\
0  3
  /\
 2 4
```

Preorder

Root → Left → Right

7, 5, 1, 0, 3, 2, 4, 6, 8, 9

Inorder

Left → Root → Right

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Postorder

Left → Right → Root

0, 2, 4, 3, 1, 6, 5, 9, 8, 7

